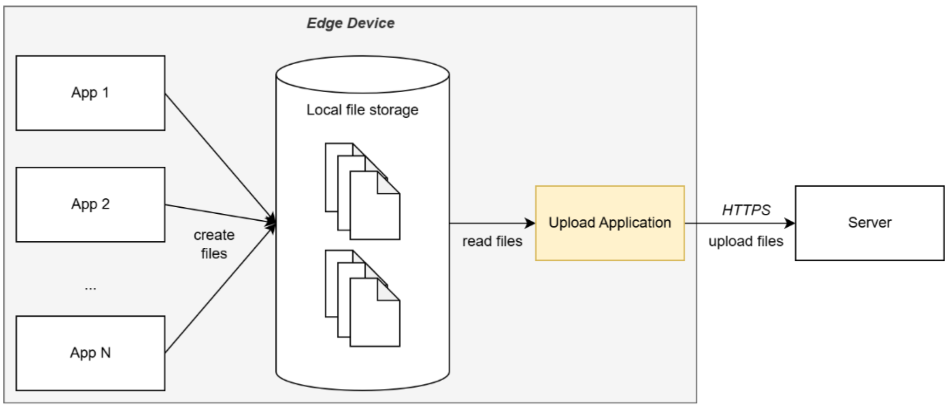
Exercise

Introduction

Our edge device produces data that must be uploaded to a server. To not lose the data in case the server is not available, the data is stored in small chunks as files on the device. New files shall be uploaded to the server as soon as they are available and deleted once they are successfully uploaded.



Tasks

Your task is to design the upload application.

1. Taking into account, that the edge device is a medical device handling sensitive patient data, which requirements do you expect?

2. Draft a concept for the upload application that fulfils these requirements.

3. Provide a minimal prototype/proof-of-concept implementation for the upload application. It should show the basic functionality, does not need to fulfil all requirements and can ignore error handling and corner cases.

Constraints

- The edge device runs a **GNU/Linux** operating system on an arm64 CPU.
- The server interface is **HTTPS** and supports
  - **HEAD** to retrieve information about (potentially) existing files,
  - **GET** to download existing files,
  - **POST** to upload a new file**,**
  - **PUT** to replace an existing file,
  - **PATCH** to continue an upload,
  - **DELETE** to delete a file.